# Stroke Management

## CLINICAL REFERENCE

KAISER PERMANENTE®
Medical Care Program,
Southern California

This Clinical Reference (CR) was developed to assist Primary Care physicians and other health professionals in the management of stroke in adults. It is adapted from the American Heart Association/American Stroke Association (AHA/ASA) 2013 Guidelines for the Early Management of Patients With Acute Ischemic Stroke *and* AHA/ASA 2010 Guidelines for the Prevention of Stroke in Patients With Stroke or Transient Ischemic Attack. The CR addresses selected interventions related to stroke care and is not intended as a comprehensive guide to stroke management. The recommendations were developed with input from the Southern California Permanente Medical Group's Stroke Clinical Reference Development Team.

## ISCHEMIC STROKE–ACUTE MANAGEMENT

### TISSUE PLASMINOGEN ACTIVATOR (TPA)

- **Intravenous administration of tissue plasminogen activator (tPA) is recommended for all adults presenting within 3 hours of acute ischemic stroke symptom onset** (see "Indications for tPA" checklist on pg. 4 for inclusion/exclusion criteria, and "tPA Dosing Nomogram" on pg. 3).
- Intra-arterial administration of tPA (IA tPA) is an ***option* within 6 hours** of symptom onset, and if there is a contraindication to administration of tPA intravenously, such as recent surgery. There is no recommendation on whether IA tPA should be used on patients whose moderate-to-severe stroke persists after IV tPA administration.
- **tPA may be administered up to 4.5 hours after stroke symptom onset with the following *additional exclusion criteria*:**
  - age older than 80 years
  - on oral anticoagulants
  - NIHSS >25 (use caution for NIHSS >22)
  - history of diabetes with stroke

For the 3- to 4.5-hour window, all patients receiving an oral anticoagulant are excluded regardless of their international normalized ratio.

In patients eligible for intravenous rtPA, benefit of therapy is time dependent, and treatment should be initiated as quickly as possible. **The ideal door-to-needle time (time of bolus administration) is within 60 minutes from hospital arrival.**

Written consent is not necessary before administration of tPA for treatment of stroke. However, a full discussion of the potential risks and benefits of treatment with tPA with the family and the patient is recommended, if possible.

### ANTITHROMBOTICS IN THE PRESENCE OF TPA CONTRAINDICATIONS

- **For adults in whom tPA is contraindicated, aspirin therapy is recommended as soon as possible, but at least within 48 hours of stroke onset, as well as at discharge.**
- **If aspirin therapy is contraindicated or not tolerated, or stroke occurred while patient was taking aspirin, then clopidogrel is an option.**

There is no recommendation for antiplatelet therapies, other than aspirin, for use in patients with acute ischemic stroke. Although aspirin plus dypyridimole has been shown to be superior to aspirin alone in some secondary prevention trials, there is a lack of evidence that aspirin plus dipyridamole is more effective than aspirin alone in the acute setting.

- **Anticoagulant therapy is not recommended for patients with acute ischemic stroke.**

### ACUTE BLOOD PRESSURE MANAGEMENT

- **Patients who have elevated blood pressure (BP) and are otherwise eligible for intravenous tPA should have their BP carefully lowered so that their systolic BP is <185 mm Hg and their diastolic BP is <110 mm Hg before tPA is initiated.**

If medications are given to lower blood pressure, ensure that the blood pressure is stabilized at the lower level before beginning treatment with tPA, and maintained below 180/105 mm Hg for at least the first 24 hours after tPA treatment.

- **In patients with markedly elevated blood pressure who do not receive tPA, a reasonable goal is to lower blood pressure by 15% during the first 24 hours after onset of stroke.**

The level of blood pressure that would mandate such treatment is not known, but consensus exists that medications are not recommended unless the systolic BP is >220 mm Hg or diastolic BP is >120 mm Hg.

| Table 1. Potential Approaches to Arterial Hypertension in Acute Ischemic Stroke Patients Who Are Candidates for Acute Reperfusion Therapy |
|---|
| Patient otherwise eligible for acute reperfusion therapy except that BP is >185/110 mm Hg:<br>• Labetalol 10–20 mg IV over 1–2 minutes, may repeat 1 time; or<br>• Nicardipine 5 mg/h IV, titrate up by 2.5 mg/h every 5–15 minutes, maximum 15 mg/h; when desired BP reached, adjust to maintain proper BP limits; or<br>• Clevidipine 1 - 2 mg/h IV, initially. Rapidly up-titrate by doubling the rate every 90 seconds until approaching target BP; thereafter, increase at 5 - 10 minute intervals by less than twice the rate; maximum rate, 16 mg/h<br>• Other agents (hydralazine, enalaprilat, etc.) may be considered, when appropriate |
| If BP is not maintained at or below 185/110 mm Hg, do not administer rtPA. |
| Management of BP during and after rtPA or other acute reperfusion therapy to maintain BP at or below 180/105 mm Hg:<br>• Monitor BP every 15 minutes for 2 hours from the start of rtPA therapy, then every 30 minutes for 6 hours, and then every hour for 16 hours |
| If systolic BP >180-230 mm Hg or diastolic BP >105-120 mm Hg:<br>• Labetalol 10 mg IV followed by continuous IV infusion 2-8 mg/min; or<br>• Nicardipine 5 mg/h IV, titrate up to desired effect by 2.5 mg/h every 5-15 minutes, maximum 15 mg/h |
| If BP not controlled or diastolic BP >140 mm Hg, consider IV sodium nitroprusside |
| BP=blood pressure; IV=intravenously; rtPA=recombinant fissue-type plasminogen activator.<br>Source: AHA/ASA 2013 Guidelines for the Early Management of Patients with Acute Ischemic Stroke |

### DEEP VEIN THROMBOSIS (DVT) PROPHYLAXIS FOR ISCHEMIC STROKE

- **For nonambulatory patients with ischemic stroke who are not at risk of bleeding, low molecular weight heparin is an option starting within 48 hours, to prevent DVT.**
- **For nonambulatory patients with ischemic stroke, early mobilization and use of intermittent pneumatic compression stockings are recommended within 48 hours or sooner, if possible.**
- **For nonambulatory patients with ischemic stroke, use of graduated compression stockings are not recommended to prevent DVT.**

# INTRACRANIAL HEMORRHAGIC (ICH) STROKE–ACUTE MANAGEMENT

### MEDICAL TREATMENT FOR ICH

- **Discontinuation of oral anticoagulant therapy is recommended for patients with oral anticoagulant -related spontaneous intracranial hemorrhage.**
- **In patients with oral anticoagulant-associated spontaneous intracranial hemorrhage, a combination of vitamin K and fresh frozen plasma (FFP) is recommended to reverse the effects of anticoagulant therapy.**
- **Three-factor Prothrombin Complex Concentrate (PCC) has not shown improved outcomes compared with FFP but may have fewer complications compared with FFP and is reasonable to consider as an alternative to FFP.** One four-factor PCC was recently FDA-approved for warfarin reversal in patients with urgent warfarin-related bleeding, but this product is still under evaluation for use in this population.
- **Recombinant factor VIIa and factor IX are not recommended for patients with intracranial hemorrhage.**

### BLOOD PRESSURE MANAGEMENT

- **In patients with a systolic BP of 150 to 220 mm Hg, acute lowering of systolic BP to 140 mm Hg is an option.**

### DEEP VEIN THROMBOSIS (DVT) PROPHYLAXIS

- **For nonambulatory patients with stroke due to spontaneous (non-traumatic) intracranial hemorrhage who are at low risk of re-bleeding, based on serial imaging exams, subcutaneous low-dose, low molecular weight heparin is an option 48 hours after ICH to prevent DVT.**
- **For nonambulatory patients with stroke due to spontaneous (non-traumatic) intracranial hemorrhage use of intermittent pneumatic compression stockings are recommended to prevent DVT.**

# SECONDARY STROKE PREVENTION

### ANTITHROMBOTIC THERAPY

#### *Stroke Due to a Noncardioembolic Source*

- **Antiplatelet therapy (aspirin, aspirin plus extended release dipyridamole or clopidogrel) is recommended for secondary prevention of noncardioembolic stroke.**
- **For patients who are allergic or intolerant to aspirin, clopidogrel is a treatment option.**
- **The addition of aspirin to clopidogrel is not recommended for secondary prevention of stroke.**

Although aspirin, aspirin plus extended-release dipyridamole and clopidogrel are acceptable first-line options for secondary stroke prevention, there is some evidence to suggest that the compound containing aspirin plus dipyridamole is more effective than aspirin alone for reducing risk of stroke recurrence.

- **For patients who have an ischemic stroke while taking aspirin, there is no evidence that increasing the dose of aspirin or changing to another antiplatelet is beneficial.**
- **Anticoagulant therapy is not recommended for patients with previous stroke if there is no obvious cardiac cause.**

#### *Stroke Due to a Cardioembolic Source*

- For patients with ischemic stroke or TIA, it is an ***option* to use transesophageal echocardiography** to detect cardiac sources of embolism, especially in patients with underlying cardiac disease.
- **In patients with ischemic stroke and nonvalvular atrial fibrillation, anticoagulation therapy, with adjusted-dose warfarin (INR 2-3, target 2.5), is recommended to prevent recurrent stroke.**

Occasionally, the management of warfarin anticoagulation may be too challenging to continue. For example, with patients who are adherent to warfarin therapy but unable to adequately keep INRs within the therapeutic range, patients who reside distant from the lab, who travel frequently, or otherwise have difficulty with INR monitoring or patients who are intolerant to warfarin, a decision is made to stop warfarin therapy.

In such cases when anticoagulation with warfarin is not desirable, **the newer target-specific oral anticoagulants (TSOAs)–dabigatran, apixaban or rivaroxaban–are an *option* to prevent recurrent stroke**. Although aspirin is often considered when the use of warfarin is not entertained, the AVERROES trial has demonstrated that TSOAs, specifically apixaban, are superior to aspirin for reducing stroke, systemic embolism without an increase in bleeding risk among atrial fibrillation patients at high risk of recurrent stroke.

- **Warfarin therapy is an *option* to prevent recurrent stroke in patients with:**
  - ischemic stroke and mechanical heart valves
  - ischemic stroke caused by an acute MI, and LV mural thrombus identified by echocardiography or other form of cardiac imaging
  - ischemic stroke and rheumatic mitral valve disease, with or without atrial fibrillation--anticoagulation with adjusted-dose warfarin at a target INR of 2.5 (range, 2.0 to 3.0)
- **For patients with ischemic stroke and dilated cardiomyopathy, either warfarin (INR 2.0 to 3.0) or antiplatelet therapy are options to prevent recurrent stroke.**
- **For patients with ischemic stroke and established cardiovascular disease (CAD) who are receiving warfarin for thromboembolic prophylaxis, low-dose aspirin (81 mg/day) is *conditionally* recommended.**

# SECONDARY STROKE PREVENTION (cont.)

Tailor the assessment of balance between benefits and harms to the individual patient's preferences and clinical circumstances. Low-dose aspirin (81 mg/day) is recommended as the risk of bleeding increases with higher aspirin dose.

#### *Stroke Due to Patent Foramen Ovale and/or Atrial Septal Aneurysm*

- **In patients with ischemic stroke and a patent foramen ovale, antiplatelet therapy is an option to prevent recurrent stroke.**
- **For high-risk patients with ischemic stroke and a patent foramen ovale who have other indications for oral anticoagulation, such as an underlying hypercoagulable state or evidence of venous thrombosis, warfarin is an option.**

There is no recommendation regarding the closure (surgical or transcatheter) of patent foramen ovale to prevent recurrent stroke in patients with ischemic stroke or atrial septal aneurysm. Similarly, there is no recommendation regarding the use of aspirin or warfarin to prevent recurrent stroke in patients with atrial septal aneurysm.

## STATIN THERAPY

- **Statin therapy is recommended for patients with ischemic stroke or transient ischemic attack.**
- **Initiate statins at a dose sufficient to reduce LDL-C to <100 mg/dL and by at least 30-40%. Treatment is recommended even if baseline LDL-C is <100 mg/dL.**
- **For patients with stroke or TIA** *and* **established CAD, an LDL-C goal of <70 mg/dL is optional.**

One study in patients with acute coronary syndrome supports an LDL-C goal of <70 mg/dL; another suggests an LDL-C goal of <80mg/dL may be appropriate for stable CAD patients.

Based on the balance of costs/harms vs. benefits, LDL-C goal of <70 mg/dL is optional. Refer to KPSC Dyslipidemia Guideline (cl.kp.org/pkc/scal/cpg/cpg/html/Dyslipid.html) for lipid step-care therapy.

## BLOOD PRESSURE MANAGEMENT

- **Combination therapy with a thiazide diuretic and ACE-inhibitor is recommended as initial treatment for post-stroke or post-TIA patients with hypertension or prehypertension.** The choice of specific drugs and targets should be individualized on the basis of pharmacological properties, mechanism of action, and consideration of specific patient characteristics for which specific agents are probably indicated (e.g., extracranial cerebrovascular occlusive disease, renal impairment, cardiac disease, and diabetes).
- **When treating an individual with a prior diagnosis of stroke (excluding subarachnoid hemorrhage, subdural hematoma and post-traumatic stroke), the target BP for hypertension is ≤139/89.**

## CAROTID ENDARTERECTOMY

- **Carotid endarterectomy (CEA) is recommended for ischemic stroke patients with 70-99% stenosis of the symptomatic carotid artery with near-occlusion.**
- CEA is an **option** for ischemic stroke patients with 50-69% stenosis of the symptomatic carotid artery.

When the degree of stenosis is <50%, there is no indication for carotid revascularization.

When CEA is indicated for patients with TIA or stroke, surgery within 2 weeks is reasonable rather than delaying surgery if there are no contraindications to early revascularization.

Carotid angioplasty and stenting (CAS) is indicated as an alternative to CEA for symptomatic patients at average or low risk of complications associated with endovascular intervention when the diameter of the lumen of the internal carotid artery is reduced by >70% by noninvasive imaging or >50% by catheter angiography.

## tPA Dosing Nomogram* in Treatment of Acute Ischemic Stroke**

| USE 50 mg VIALS | | | | |
|---|---|---|---|---|
| WEIGHT | BOLUS DOSE IV Push over 1 min | INFUSION DOSE over 60 minutes | | TOTAL DOSE wt (kg) x 0.9mg/kg |
| | | Dose | Rate | |
| 60 lbs = 27 kg | 2 mg = 2 mL | 22 mg | 22 mL/hr | 24 mg |
| 65 lbs = 30 kg | 3 mg = 3 mL | 24 mg | 24 mL/hr | 27 mg |
| 70 lbs = 32 kg | 3 mg = 3 mL | 26 mg | 26 mL/hr | 29 mg |
| 75 lbs = 34 kg | 3 mg = 3 mL | 28 mg | 28 mL/hr | 31 mg |
| 80 lbs = 36 kg | 3 mg = 3 mL | 29 mg | 29 mL/hr | 32 mg |
| 85 lbs = 39 kg | 4 mg = 4 mL | 31 mg | 31 mL/hr | 35 mg |
| 90 lbs = 41 kg | 4 mg = 4 mL | 33 mg | 33 mL/hr | 37 mg |
| 95 lbs = 43 kg | 4 mg = 4 mL | 35 mg | 35 mL/hr | 39 mg |
| 100 lbs = 45 kg | 4 mg = 4 mL | 37 mg | 37 mL/hr | 41 mg |
| 105 lbs = 48 kg | 4 mg = 4 mL | 39 mg | 39 mL/hr | 43 mg |
| 110 lbs = 50 kg | 5 mg = 5 mL | 40 mg | 40 mL/hr | 45 mg |
| 115 lbs = 52 kg | 5 mg = 5 mL | 42 mg | 42 mL/hr | 47 mg |
| 120 lbs = 55 kg | 5 mg = 5 mL | 45 mg | 45 mL/hr | 50 mg |

| USE 100 mg VIALS | | | | |
|---|---|---|---|---|
| WEIGHT | BOLUS DOSE IV Push over 1 min | INFUSION DOSE over 60 minutes | | TOTAL DOSE wt (kg) x 0.9mg/kg |
| | | Dose | Rate | |
| 125 lbs = 57 kg | 5 mg = 5 mL | 46 mg | 46 mL/hr | 51 mg |
| 130 lbs = 59 kg | 5 mg = 5 mL | 48 mg | 48 mL/hr | 53 mg |
| 135 lbs = 61 kg | 6 mg = 6 mL | 49 mg | 49 mL/hr | 55 mg |
| 140 lbs = 64 kg | 6 mg = 6 mL | 52 mg | 52 mL/hr | 58 mg |
| 145 lbs = 66 kg | 6 mg = 6 mL | 53 mg | 53 mL/hr | 59 mg |
| 150 lbs = 68 kg | 6 mg = 6 mL | 55 mg | 55 mL/hr | 61 mg |
| 155 lbs = 70 kg | 6 mg = 6 mL | 57 mg | 57 mL/hr | 63 mg |
| 160 lbs = 73 kg | 7 mg = 7 mL | 59 mg | 59 mL/hr | 66 mg |
| 165 lbs = 75 kg | 7 mg = 7 mL | 61 mg | 61 mL/hr | 68 mg |
| 170 lbs = 77 kg | 7 mg = 7 mL | 63 mg | 63 mL/hr | 70 mg |
| 175 lbs = 80 kg | 7 mg = 7 mL | 65 mg | 65 mL/hr | 72 mg |
| 180 lbs = 82 kg | 7 mg = 7 mL | 67 mg | 67 mL/hr | 74 mg |
| 185 lbs = 84 kg | 8 mg = 8 mL | 68 mg | 68 mL/hr | 76 mg |
| 190 lbs = 86 kg | 8 mg = 8 mL | 69 mg | 69 mL/hr | 77 mg |
| 195 lbs = 89 kg | 8 mg = 8 mL | 72 mg | 72 mL/hr | 80 mg |
| 200 lbs = 91 kg | 8 mg = 8 mL | 74 mg | 74 mL/hr | 82 mg |
| 205 lbs = 93 kg | 8 mg = 8 mL | 76 mg | 76 mL/hr | 84 mg |
| 210 lbs = 95 kg | 9 mg = 9 mL | 77 mg | 77 mL/hr | 86 mg |
| 215 lbs = 98 kg | 9 mg = 9 mL | 79 mg | 79 mL/hr | 88 mg |
| 220 lbs = 100 kg | 9 mg = 9 mL | 81 mg | 81 mL/hr | 90 mg |
| **For Weights Greater Than 220 lbs (100 kg)** | **9 mg = 9 mL** | **81 mg** | **81 mL/hr** | **90 mg** |

*tPA is administered over 60 minutes, with 10% of the total dose administered as an IV bolus over 1 minute and the remaining 90% of the dose as continuous infusion over 60 minutes.

**Adapted from the Woodland Hills Medical Center Pharmacy and Therapeutics Committee.

# Checklist – Indications for tPA

| INCLUSION CRITERIA (all answers must be YES to be treated) | YES | | NO |
|---|---|---|---|
| • Ischemic stroke in any vascular territory, large or small vessel | | | |
| • Age at least 18 years<br>o Age 18-80 for tPA administration 3 to 4.5 hours after onset of stroke symptoms | | | |
| • Ability to establish a definite time of onset of stroke symptoms, defined by time last known well | | | |
| • Ability to begin administration of rt-PA within 3 hours of onset of stroke symptoms<br>o Consult Neurology for administration of tPA 3 to 4.5 hours after onset of stroke symptoms; additional exclusion criteria* apply | | | |
| **ABSOLUTE EXCLUSION CRITERIA (all answers must be NO to be treated)**<br>**(Patients for whom any of the following cannot be determined should also be excluded)** | **YES** | | **NO** |
| • Stroke or serious head trauma or intracranial surgery within the past 3 months** | | | |
| • CT brain scan evaluated by Neurologist or Radiologist showing evidence of any type of hemorrhage, tumor or early signs of cerebral edema, including but not limited to the following: | | | |
| o High-density lesion consistent with hemorrhage | | | |
| o Significant mass effect such as effacement of ventricles or midline shift | | | |
| o Sulcal effacement in more than one-third (1/3) of middle cerebral artery (MCA) territory with loss of the gray-white matter differentiation | | | |
| • High suspicion of subarachnoid hemorrhage despite normal CT brain scan | | | |
| • Any history of intracranial hemorrhage, subarachnoid hemorrhage, arteriovenous malformation, or intracranial aneurysm | | | |
| • Sustained (on repeated measurements) systolic BP >185 or diastolic BP >110 mm Hg | | | |
| • Blood glucose concentration of <50 or >400 mg/dL | | | |
| • Known bleeding diathesis | | | |
| **RELATIVE EXCLUSION CRITERIA**<br>**(Consider excluding patients for whom any of the following cannot be determined)** | **YES** | | **NO** |
| • Seizure at onset of stroke | | | |
| • Rapidly improving symptoms | | | |
| • Isolated, mild neurological deficits, such as ataxia alone, sensory loss alone, dysarthria alone, or minimal weakness** | | | |
| • Altered mental state due to coma, stupor or alcohol or drug abuse | | | |
| • Elevated PTT ≥1.5*** | | | |
| • INR >1.7*** | | | |
| • Platelet count <100,000/mm$^3$*** | | | |
| • Gastrointestinal or urinary tract hemorrhage within the past 21 days | | | |
| • Major surgery within the past 14 days | | | |
| • A-line at any non-compressible site or biopsy within the past 7 days | | | |
| • Lumbar puncture within the past 7 days | | | |
| • Pericarditis, endocarditis, septic embolism to the brain | | | |
| • Inflammatory bowel disease | | | |
| • Myocardial infarction within the past 3 months | | | |
| • Currently taking direct thrombin inhibitors or direct factor Xa inhibitors (unless PTT, PT, direct factor Xa or INR are normal, or the patient has not received a dose of these agents for >2 days, assuming normal renal metabolizing function) | | | |
| Pregnancy is a relative but not absolute contraindication. Treat pregnant stroke patients on a case-by-case basis. | | | |
| Consider a guaiac test (FOBT) as a positive result might present a contraindication in some patients. Treat on a case-by-case basis. | | | |

*Additional exclusion criteria for tPA administration 3 to 4.5 hours after stroke onset: age > 80 years; on oral anticoagulants (regardless of INR); NIHSS >25 (use caution for NIHSS >22); history of diabetes with stroke.

**Use of intravenous fibrinolysis in patients with conditions of mild stroke deficits, rapidly improving stroke symptoms, major surgery in the preceding 3 months, and recent myocardial infarction may be considered, and potential increased risk should be weighed against the anticipated benefits.

***Stat blood work should be obtained immediately upon arrival, but administration of IV tPa should not be delayed while waiting for PT, PTT, INR or in patients without known bleeding problems; infusion can be stopped if these are abnormal.

# Stroke Management: Medication Table

| | | |
|---|---|---|
| **Fibrinolytic Agent** | | [see Tissue Plasminogen Activator section on page 1] |
| **Alteplase, rh-tPA**<br>Activase® | | For patients presenting within 3 hours of symptom onset, treatment with intravenous tissue plasminogen activator (tPA) is recommended. tPA may be administered up to 4.5 hours after stroke symptom onset with the additional exclusion criteria listed on page 1. (See Indications for tPA on Page 3) |
| **Antiplatelet Agents** | | |
| **Aspirin, (ASA)**<br>Various® | **Dose: 81-325 mg orally daily** | Aspirin therapy is recommended as soon as possible but at least within 48 hours of stroke onset in acute ischemic stroke patients and at discharge.<br>Low-dose aspirin (81 mg/day) is conditionally recommended for most patients with history of stroke and established CAD who are receiving warfarin for thromboprophylaxis.<br>The balance between benefits and harms requires individualized assessment and should be tailored to the individual patient's preferences and clinical circumstances. Low-dose aspirin (81 mg/day) is recommended because the risk of bleeding increases with higher aspirin dose.<br>For patients with non-valvular AF in whom anticoagulation is contraindicated, aspirin 325 mg/d is recommended. |
| **Clopidogrel**<br>Plavix® | **Dose: 75 mg orally daily** | For patients with acute ischemic stroke who are allergic or intolerant to aspirin, clopidogrel is an option. Elective surgery; discontinue use 5 days prior if antiplatelet effect is not desired.<br>Premature discontinuation of therapy in percutaneous coronary intervention patients may increase risk of stent thrombosis, myocardial infarction, and death. |
| **ASA/MR-Dipyridamole**<br>*B Aggrenox® | **Dose: 25/200 mg orally BID** | |
| | **Potential Adverse Effects**<br>Hemorrhage<br>GI distress, rash, nausea<br>Very rarely, thrombocytopenia<br>Headache (dipyridamole)<br>**Follow Up Labs**<br>Hemoglobin/Hematocrit, CBC w/Diff, platelet count | Precautions<br>Combination of aspirin and clopidogrel in patients with recent TIA or stroke that are at risk for recurrent ischemic events has not been shown to be more effective than clopidogrel alone; the combination has been shown to increase major bleeding. |
| *B = available as Brand-name only | | |
| **Antithrombins** | | |
| **Unfractionated Heparin (UFH)**<br>Various® | **Dose: 5000 units SC BID-TID** | Anticoagulant therapy is not recommended for patients in the acute stroke setting, because of evidence for the increased risk of hemorrhage. Consider SC LMWH or low-dose UFH only for patients at high risk of PE or DVT. **Low-dose UFH is preferred in those with kidney disease.** |
| **Enoxaparin**<br>Lovenox® | **Dose: 30 mg SC BID or 40 mg SC once daily** | Consider SC LMWH or low-dose UFH only for patients at high risk of PE or DVT. |
| **Fondaparinux**<br>Arixtra® | **Dose: 2.5 mg SC once daily** | For patients with ischemic stroke who are non-ambulatory and are not taking ASA or at risk of bleeding, heparin is recommended within 48 hours or sooner if possible.<br>50% enoxaparin dose reduction is required if GFR<30.<br>Thromboprophylaxis with fondaparinux is contraindicated for weight <50 kg and GFR<30. |
| | **Potential Adverse Effects**<br>Minor to severe Hemorrhage<br>**Follow Up Labs**<br>Monitored using aPTT, anti-Factor Xa as appropriate.<br>Follow Hemoglobin/Hematocrit, platelet count | Precautions<br>During or immediately following spinal tap, spinal anesthesia, brain, spinal or ophthalmological surgery; increased risk of hemorrhage exists.<br>Pregnancy (risk:benefit). |
| **Vitamin K Antagonist Anticoagulant** | | |
| **Warfarin**<br>Coumadin® | **Start: 0.5-10 mg orally daily**<br>**Max daily: 5 mg orally daily**<br>**Dose according to INR**<br>**Target INR: 2.5 (range 2.0 to 3.0)** | **Consider KP Outpatient Anticoagulation Management Program.** In patients with ischemic stroke and nonvalvular AF, anticoagulation with adjusted-dose warfarin (INR 2-3, target 2.5) is recommended to prevent recurrent stroke. For patients with ischemic stroke who have rheumatic mitral valve disease, whether or not AF is present, anticoagulation with adjusted-dose warfarin at a target INR of 2.5 (range, 2.0 to 3.0) is an option to prevent recurrent stroke.<br>Low-dose aspirin (81 mg/day) is conditionally recommended for most patients with history of stroke and established CAD who are receiving warfarin for thromboprophylaxis.<br>The balance between benefits and harms requires individualized assessment and should be tailored to the individual patient's preferences and clinical circumstances. Low-dose aspirin (81 mg/day) is recommended because the risk of bleeding increases with higher aspirin dose. |
| | **Potential Adverse Effects**<br>Minor-to-severe hemorrhage<br>**Follow Up Labs**<br>Routine INR surveillance; Hemoglobin/Hematocrit | **Precautions**<br>The desired effect of the therapeutic use of warfarin is anticoagulation. However, undesirable manifestations have been reported including increased warfarin sensitivity, excessive prothrombin times, eosinophilia, and anemia.<br>Multiple moderate-to-severe drug-drug interactions.<br>Pregnancy (birth defects). |

# Stroke Management: Medication Table (Continued)

## Antihypertensive Agents

[Refer to the KPSC Hypertension Clinical Practice Guidelines]

### ACE Inhibitors

| | | |
|---|---|---|
| **Lisinopril**<br>Prinivil,® Zestril | **Start: 10 mg orally daily**<br>**Max daily: 40 mg** | If CVD or microalbuminuria; or if age ≥55 years and ≥1 other cardiovascular risk factors (refer to the KPSC Hypertension Clinical Practice Guideline at http://cl.kp.org/pkc/scal/cpg/cpg/html/HTN.html). |
| **Lisinopril/Hydrochlorothiazide**<br>Prinzide® | **Start: 10/12.5 mg orally daily**<br>**Max daily: 20/25 mg** | |
| **Benazepril**<br>Lotensin®I | **Start: 10mg orally daily**<br>**Max daily: 80 mg** | |
| | **Potential Adverse Effects**<br>Cough, angioedema, orthostatic hypotension, hyperkalemia, or creatinine rise >20-30%<br>**Follow Up Labs**<br>Baseline creatinine and K+ and at 1-3 weeks after initiation or dosage increase<br>If creatinine rises >20-30%, consider decreasing or discontinuing dose | **Precautions**<br>K+ >5.5 mEq/L creatinine >1.5 mg/dL<br>(Heart Failure: creatinine >3.0 mg/dL)<br>SBP<95 or symptomatic hypotension<br>ACE-I induced angioedema<br>K+ sparing diuretics<br>Pregnancy (birth defects) |

### Diuretics

| | | |
|---|---|---|
| **Hydrochlorothiazide**<br>HydroDiuril® | **Start: 12.5-25 mg orally daily**<br>**Max daily: 25 mg** | |
| **Chlorthalidone**<br>Hygroton® | **Start: 12.5-25 mg orally daily**<br>**Max daily: 25 mg** | |
| | **Potential Adverse Effects**<br>Hypokalemia, hyponatremia, rash<br>**Follow Up Labs**<br>K+, electrolytes: 1 week after initiation and annually | **Precautions**<br>Hypokalemia<br>Pregnancy (blood pressure) |

### Beta-Blockers

| | | |
|---|---|---|
| **Atenolol**<br>Tenormin® | **Start: 25-50 mg orally daily**<br>**Max daily: 100 mg** | Use in diabetes patients with CAD and/or HTN<br>Use carvedilol, bisoprolol, or metoprolol succinate with heart failure |
| **Metoprolol**<br>Lopressor® | **Start: 25-50 mg orally BID**<br>**Max: 100 mg BID** | |
| **Atenolol/Chlorthalidone**<br>Tenoretic® | **Start: 50/25 mg orally daily**<br>**Max daily: 100/25 mg** | |
| **Bisoprolol**<br>Zebeta® | **Start: 2.5-5 mg orally daily**<br>**Max daily: 20 mg** | |
| **Metoprolol succinate**<br>Toprol XL® | **Start: 12.5-25 mg orally daily**<br>**Max daily: 200 mg** | |
| | **Potential Adverse Effects**<br>Masked hypoglycemia, orthostasis, dizziness, asthma, worsening PVD | **Precautions:** Heart rate <50, SBP <95; Uncontrolled severe asthma; Pregnancy (blood pressure) |

## Lipid-Lowering Therapy

[Refer to the KPSC Dyslipidemia Management Clinical Practice Guideline]

### HMG CoA Reductase Inhibitors (Statins)

| | | |
|---|---|---|
| **Atorvastatin**<br>Lipitor® | **Start: 10-20 mg**<br>**Max daily: 80 mg**** | Target LDL <100 mg/dL; initiate statins at a dose expected to reduce LDL-C by 30 to 40 percent. Treatment is recommended even if baseline LDL-C is <100 mg/dL.<br>Renal Function: Adjustment of atorvastatin dosing is not necessary for impaired renal function. However, when drug interactions require the substitution of either pravastatin or rosuvastatin-NF, dosing of these two statins should be based on renal function. No dosage adjustment is required for patients with eGFR >30 mL/min. For patients not on hemodialysis with severe renal insufficiency (i.e., with eGFR between 15 and 30 mL/min), the maximum recommended dose of pravastatin is 40 mg per day, and for rosuvastatin-NF the daily dose should not exceed 10 mg. For patients with stroke/TIA who have diabetes or CAD, an LDL-C goal of <70 mg/dL is optional.<br>Inadequate LDL lowering response to statin can be enhanced by combination therapy (refer to KPSC Dyslipidemia Management Guideline at http://cl.kp.org/pkc/scal/cpg/cpg/html/Dyslipid.html).<br>Pravastatin is not significantly metabolized by the CYP3A4 pathway, unlike lovastatin and simvastatin. In patients who commonly require medications that strongly inhibit CYP3A4, pravastatin may be selected to minimize the risk of myotoxicity resulting from CYP3A4-mediated drug interactions. |
| **Simvastatin**<br>Zocor® | **Start: 20-40 mg**<br>**Max daily: 80 mg** | |
| **Lovastatin**<br>Mevacor® | **Start: 40-80 mg**<br>**Max daily: 80 mg** | |
| **Pravastatin**<br>Pravachol® | **Start: 40-80 mg**<br>**Max daily: 80 mg** | |
| **Rosuvastatin-NF**<br>*B Crestor® | **Start: 5-10 mg**<br>**Max daily: 40 mg**<br>**Initiating at, or up-titrating to 80 mg daily, is not recommended due to risk of myopathy. It is acceptable for patients to remain on simvastatin 80 mg daily if they have completed one year of therapy at that dose without muscle complaints. | |
| NF=Non-formulary<br>*B = available as Brand-name only | **Potential Adverse Effects**<br>Myalgia, rhabdomyolysis (rare), liver disease (rare)<br>**Follow Up Labs**<br>Lipid panel/ALT 6-8 weeks after initiation and after each dosage increase<br>Lipid panel/ALT/serum creatinine when clinically appropriate | **Precautions**<br>Use caution if statins are used together with niacins or fibrates due to the increased risk of severe myopathy, rhabdomyolysis, and acute kidney disease. Statin dosage should be reduced.<br>Stop the statin if the ALT increases to >3 times the upper limit of normal, or if significant generalized muscle aches thought to be caused by the drug.<br>Although potentially fatal rhabdomyolysis is very rare, myopathy (myalgia, weakness, myositis) occurs more frequently, the risk for statin-induced muscle injury is increased in the elderly (especially >80 years), people with impaired renal function, and in the presence of interacting drugs. Consider reduced dose in these populations.<br>Pregnancy (birth defects) |

## References

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